Practice Sheet 2

1. Evaluate the following limits:

(i) $\lim_{z\to 1+i} \frac{z^2 - z + 1 - i}{z^2 - 2z + 2}$ (ii) $\lim_{z\to 1+i} \left\{ \frac{z - 1 - i}{z^2 - 2z + 2} \right\}^2$ (iii) $\lim_{z\to i} \frac{z^2 + 1}{z^6 + 1}$.

2. If $f(z) = \frac{2z - 1}{3z + 2}$, prove that $f'(z_0) = \lim_{h\to 0} \frac{f(z_0 + h) - f(z_0)}{h} = \frac{7}{(3z_0 + 2)^2}$ provided $z_0 \neq -\frac{2}{3}$.

3. Let $f(z) = \frac{z^2 + 4}{z - 2i}$ if $z \neq 2i$, while $f(2i) = 3 + 4i$. Is $f(z)$ continuous at $z = 2i$?

4. Find all points of discontinuity for the function $f(z) = \frac{2z - 3}{z^2 + 2z + 2}$.

5. Using the definitions, find the derivative of each function at the indicated points

(i) $f(z) = \frac{2z - i}{z + 2i}$ at $z = -i$

(ii) $f(z) = 3z^{-2}$ at $z = 1 + i$.

6. Evaluate the following limits using L' Hôpital's rule

(i) $\lim_{z\to 2i} \frac{z^2 + 4}{2z^2 + (3 - 4i)z - 6i}$ (ii) $\lim_{z\to 0} \frac{z - \sin z}{z^3}$.

7. Determine which of the following functions u are harmonic. For each harmonic function find the conjugate harmonic function v and express $u + iv$ as an analytic function of z

i) $u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2$ ii) $u(x, y) = xe^x \cos cos\, y - ye^x \sin sin\, y$

iii)

$u(x, y) = e^{-x}(x \sin sin\, y - y \cos cos\, y$